

EcoHydroGraph: Hydro–ecological temporal graph learning for sparse river water-quality reconstruction

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Abstract

River monitoring networks rarely observe every water-quality indicator at every station and month. These gaps are structured by river connectivity, watershed ecology, hydrologic conditions, and sampling schedules. We introduce **EcoHydroGraph**, a temporal river-graph model that combines a spatial graph encoder, watershed ecological features, hydroclimatic inputs, and a causal gated recurrent unit. We evaluate separate models for dissolved organic carbon (DOC), pH, and specific conductance on an audited 357-station, 654-month river-network cohort under random, temporal, and spatial missingness. In a five-seed matched comparison, EcoHydroGraph reduces DOC mean absolute error by 37.5–47.2% and specific-conductance error by 38.8–57.2% relative to the matched snapshot graph model. pH gains are smaller (0.5–3.8%) and depend on the missingness regime. Window diagnostics show that most pH and conductance gains appear by six months, whereas DOC changes little with window length. The results show that analyte-specific hydro–ecological temporal context can improve sparse water-quality reconstruction.

Keywords: river monitoring; water-quality reconstruction; graph neural network; ecological informatics; temporal memory; dissolved organic carbon

1 Introduction

Long-term river monitoring networks provide broad spatial and temporal coverage, but observations remain sparse and irregular. Stations differ in sampling frequency, programs change over time, and access or instrumentation creates gaps. These gaps are embedded in a river network, a watershed ecological context, and a changing hydrologic regime. This limits the use of raw records for carbon-cycle analysis, salinity assessment, acid–base characterization, extreme-event detection, and comparisons among watersheds.

Conventional interpolation uses local spatial or temporal similarity. This is difficult to operationalize when gaps arise from network failures, device replacement, irregular sampling, or other mechanisms that can bias downstream analysis [8, 9]. Tabular machine-learning models add environmental covariates while treating station-month records as independent samples. Graph neural networks encode relational structure [1, 2]; spatio-temporal graph models learn spatial and temporal dependencies jointly [5], and hydrological applications demonstrate their potential for prediction at unobserved locations [6]. River-temperature work also shows that the choice of spatial and temporal architecture affects robustness when geographic or climatic conditions shift [7]. For example, Choi et al. extend graph learning toward sparse water-quality monitoring and management interpretation through a chromaticity case study in the Hantan River Basin [10]. The open problem is to combine spatial graph structure, ecological context, and hydroclimatic temporal memory while testing the result under the distinct ways a monitoring network becomes incomplete.

We develop EcoHydroGraph. A spatial river-graph encoder combines station inputs, edge attributes, and ecological features at each month. A causal GRU [3] integrates the current representation with the preceding eleven months. Here causal describes the information-flow constraint that a prediction at month t receives no future-month labels; it is not a causal-inference claim. Separate models are trained for DOC, pH, and specific conductance.

Our contributions are threefold: (i) a hydro-ecological temporal graph model for sparse station-month reconstruction; (ii) evidence that temporal information has analyte- and missingness-dependent value, with large gains for DOC and specific conductance and smaller gains for pH; and (iii) a matched evaluation framework that separates random, temporal, and spatial missingness and measures the useful history length. The design isolates the value of temporal context by keeping the spatial encoder and target masks fixed between the snapshot and temporal models.

2 Materials and Methods

2.1 Cohort, targets, and inputs

The audited ST357 cohort contains 357 stream stations, 324 river-network edges, and 654 monthly time points. DOC and specific conductance use a log1p target transform; pH is standardized on the raw scale using training cells only. Monthly inputs contain temperature, discharge, visibility flags, seasonal sine/cosine terms, static station coordinates, ecological regime features, and target-value/visibility channels. The graph input contains the frozen river edge set and edge attributes. All targets are standardized after the target transform using training-label statistics. Dynamic input scaling uses training-cell locations; coordinates and ecological descriptors are standardized across the known station cohort. This setting assumes that station descriptors and contemporaneous hydrological inputs are available at prediction locations.

2.2 Missingness and visibility

We evaluate four scenarios: *random gaps*, which hide individual observed cells; *unobserved-period extrapolation*, which hides all target observations after the temporal cutoff; *observation-assisted extrapolation*, which retains a subset of post-cutoff observations as context; and *unmonitored stations*, which withholds target observations at selected stations. Role cells are intersected with each analyte’s observed-label mask. During fitting, half of the training cells are randomly retained as input context each epoch and the remainder provide the loss targets, alongside any designated context cells. Validation labels never enter the fitting or early-stopping inputs; they are used to calculate the early-stopping loss. Final inference exposes training, validation, and designated context labels, while test labels remain hidden. Each temporal window ends at the prediction month. Snapshot and temporal models use identical target masks and query cells.

2.3 EcoHydroGraph

The spatial trunk is a two-layer transport graph encoder with hidden width 64, ecological encoding, and dropout 0.1. It is applied to every monthly station graph. The resulting node representations enter a one-layer GRU with hidden width 64 and a 12-month causal window. The final recurrent representation is mapped through the prediction head to the target analyte. The snapshot baseline uses the same spatial trunk without the temporal wrapper. Figure 1 summarizes the architecture.

For station i and month t , let $x_{i,t}$ denote the monthly input and e_i the ecological descriptor. The ecological encoder is a two-layer perceptron with width 32. Its output is concatenated with $x_{i,t}$.

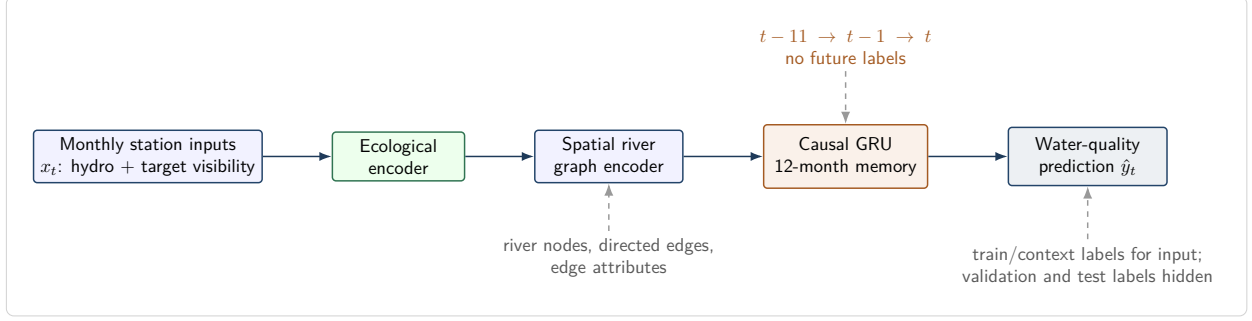


Figure 1: **EcoHydroGraph architecture.** Monthly hydroclimatic, ecological, and target-visibility inputs are encoded on the river graph. The spatial encoder is applied to the preceding eleven months and the current month, after which a causal GRU produces the current prediction.

Each spatial layer computes

$$h_{i,t}^{(\ell+1)} = \text{Dropout} \left[\text{ReLU} \left(W_0^{(\ell)} h_{i,t}^{(\ell)} + b_0^{(\ell)} + \sum_{r \in \{\text{up}, \text{down}\}} m_{i,t,r}^{(\ell)} \right) \right], \quad (1)$$

where

$$m_{i,t,r}^{(\ell)} = \frac{1}{\max(1, |\mathcal{N}_r(i)|)} \sum_{j \in \mathcal{N}_r(i)} \sigma(g_r^{(\ell)}(a_{ji})) (W_r^{(\ell)} h_{j,t}^{(\ell)} + b_r^{(\ell)}). \quad (2)$$

Here a_{ji} contains edge attributes, g_r is a learned two-layer gate network with intermediate width 16, and σ is the sigmoid function. Upstream and downstream messages have separate parameters. The gates are learned functions of static edge attributes.

The temporal module receives $[h_{i,\tau}^{(2)}, v_\tau]$ for $\tau = t - 11, \dots, t$, where v_τ indicates a valid history position. A fresh recurrent state is used for each window, and the last GRU state is passed through a scalar linear head. At the start of the record, unavailable history positions repeat the first encoded month and have $v_\tau = 0$. Thus, padding adds no distinct earlier observation.

2.4 Training and evaluation

Primary comparisons use five seeds (42–46) and the matched 10-epoch, patience-3 budget for both models. Training minimizes mean squared error on standardized targets with Adam (learning rate 10^{-3} , no weight decay) and gradient-norm clipping at 1. The checkpoint with the lowest validation loss is retained. Before inverse transformation, log-transformed predictions are clipped to the training minimum and 99.5th percentile; pH predictions are clipped to the training minimum and maximum. The primary metric is MAE on hidden query cells; RMSE, R^2 , log-space error, and train-derived Q90 tail MAE are secondary. Tails with fewer than 20 query cells are flagged unstable. Errors are averaged across seeds at each query cell before 2,000 paired station- or month-clustered bootstrap replicates. The primary uncertainty table reports station-clustered 95% intervals for the paired MAE difference; month-clustered intervals are retained in the analysis files. Relative reduction is $100(1 - \text{MAE}_{\text{EcoHydroGraph}} / \text{MAE}_{\text{snapshot}})$.

Table 1: Five-seed matched comparison. Positive reduction favors EcoHydroGraph.

Analyte	Scenario	Snapshot MAE	EcoHydroGraph MAE	Reduction (%)	Better seeds
DOC	Random gaps	4.285	2.625	38.75	5/5
DOC	Strict temporal	2.679	1.417	47.12	5/5
DOC	Context-assisted temporal	2.679	1.413	47.24	5/5
DOC	Unmonitored stations	6.198	3.876	37.47	5/5
pH	Random gaps	0.347	0.335	3.53	4/5
pH	Strict temporal	0.313	0.302	3.58	4/5
pH	Context-assisted temporal	0.312	0.300	3.77	4/5
pH	Unmonitored stations	0.286	0.284	0.51	3/5
Specific conductance	Random gaps	683.2	374.8	45.14	5/5
Specific conductance	Strict temporal	572.7	247.9	56.71	5/5
Specific conductance	Context-assisted temporal	568.9	243.5	57.19	5/5
Specific conductance	Unmonitored stations	996.4	609.4	38.84	5/5

3 Results

3.1 EcoHydroGraph improves sparse reconstruction

Table 1 gives the five-seed matched comparison. EcoHydroGraph improves DOC in all four regimes, with reductions from 37.5% in unmonitored stations to 47.2% in observation-assisted extrapolation. Specific conductance shows reductions from 38.8% in unmonitored stations to 57.2% in observation-assisted extrapolation. Both analytes improve for all five seeds in every regime. The largest gains occur in the two temporal extrapolation scenarios, where the query month is separated from observed target history.

The paired station-clustered intervals in Table 2 remain below zero for every DOC and conductance scenario. For pH, the intervals are below zero in random gaps and both temporal extrapolation scenarios, but include zero for unmonitored stations. This pattern is consistent with an analyte- and missingness-dependent temporal benefit rather than a uniform improvement.

pH shows a smaller effect: reductions range from 0.5% in unmonitored stations to 3.8% in observation-assisted extrapolation. Random gaps and both temporal extrapolation scenarios improve for four of five seeds. Reconstruction at unmonitored stations is close to parity, with the station-clustered interval including zero.

3.2 Temporal history length has analyte-specific value

The window diagnostic compares 1, 3, 6, and 12 months in the two temporal extrapolation scenarios using three matched seeds. Mean MAE decreases from 1.420 to 1.415 for DOC, from 0.362 to 0.301 for pH, and from 279.12 to 245.75 for specific conductance. Most pH and conductance improvement appears by six months, while DOC is nearly flat. We retain 12 months as the common working configuration.

3.3 Longer history matters more than one isolated channel

The reverse-history and hydro-only arms remain within 0.6% of the full temporal model in the three-seed diagnostic. The current-only control is 20.9% worse for pH and 13.8% worse for specific conductance in the two temporal extrapolation scenarios, while DOC changes by 0.5%. These results

Table 2: Station-clustered 95% bootstrap intervals for the paired MAE difference (EcoHydroGraph minus snapshot). Negative values favor EcoHydroGraph.

Analyte	Scenario	Δ MAE	95% CI
DOC	Random gaps	-1.660	[-1.894, -1.415]
DOC	Strict temporal	-1.263	[-1.722, -0.763]
DOC	Context-assisted temporal	-1.266	[-1.725, -0.740]
DOC	Unmonitored stations	-2.322	[-2.770, -1.900]
pH	Random gaps	-0.012	[-0.019, -0.007]
pH	Strict temporal	-0.011	[-0.017, -0.005]
pH	Context-assisted temporal	-0.012	[-0.018, -0.006]
pH	Unmonitored stations	-0.001	[-0.005, 0.002]
Specific conductance	Random gaps	-308.4	[-345.1, -272.8]
Specific conductance	Strict temporal	-324.8	[-396.2, -247.1]
Specific conductance	Context-assisted temporal	-325.4	[-396.9, -245.3]
Specific conductance	Unmonitored stations	-387.0	[-443.2, -321.8]

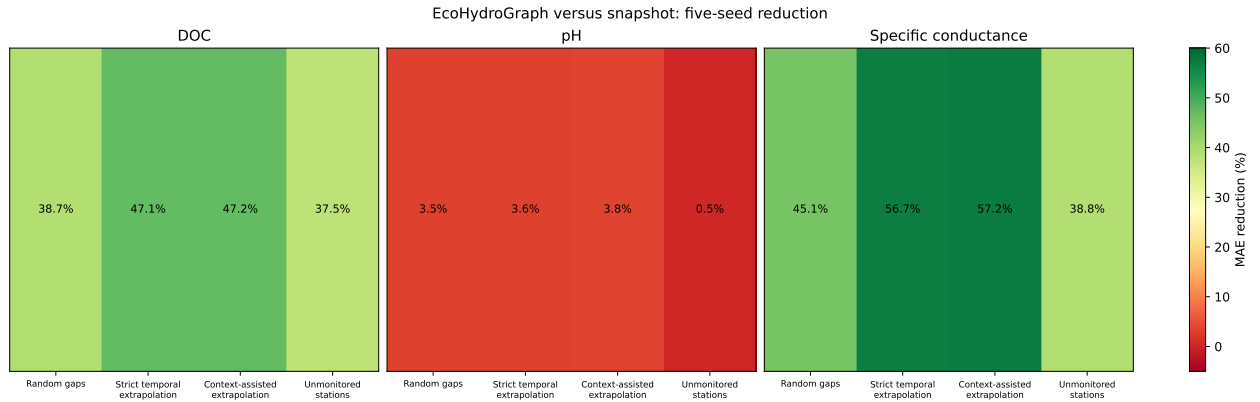


Figure 2: **Analyte- and regime-specific improvement.** Relative MAE reduction of EcoHydroGraph relative to the matched snapshot graph model.

identify the amount of temporal context as useful for pH and conductance. They do not isolate strict chronological order from the amount of history, and they do not establish that target-analyte history is the source of the gain.

3.4 Spatially heterogeneous reconstruction error

The five-seed ensemble produces a full station-month reconstruction for every analyte and regime. Figure 5 shows test-cell error under unobserved-period extrapolation by station. Error is spatially heterogeneous, and the scale differs among analytes.

Table 3: Mean MAE across the two temporal extrapolation scenarios by temporal lookback.

Analyte	1 month	3 months	6 months	12 months
DOC	1.420	1.419	1.417	1.415
pH	0.362	0.319	0.305	0.301
Specific conductance	279.12	255.19	247.72	245.75

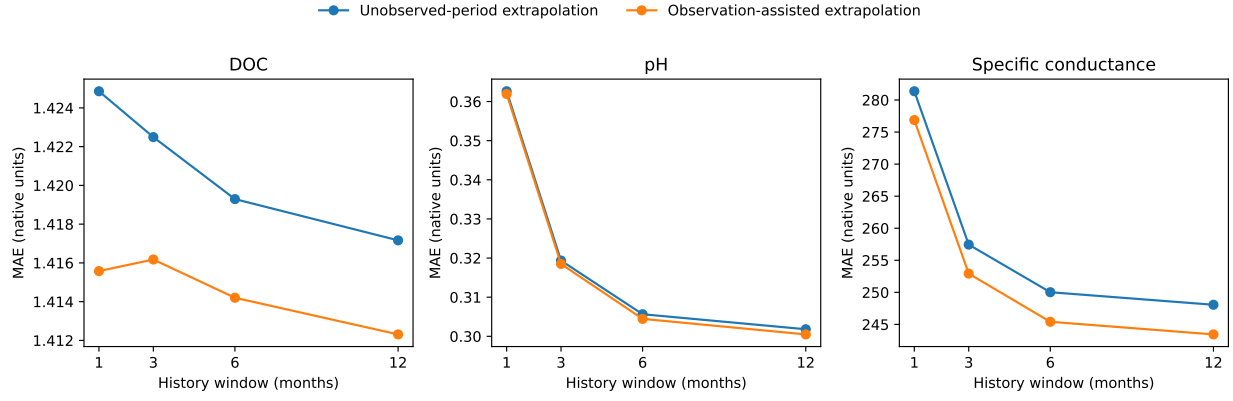


Figure 3: **Temporal window diagnostic.** Longer history improves pH and specific conductance, with diminishing gains beyond six months. DOC changes little.

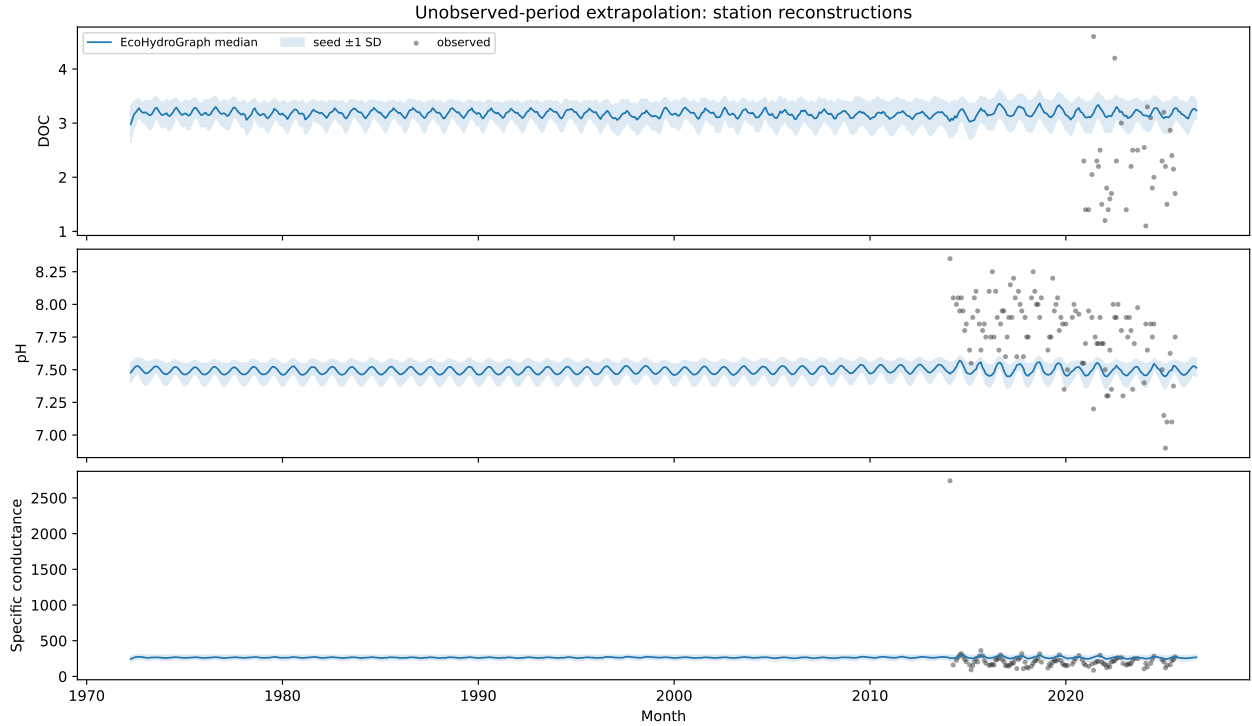


Figure 4: **Descriptive station traces under unobserved-period extrapolation.** Five-seed EcoHydroGraph median predictions, observed values, and seed dispersion for one fixed data-availability-selected station per analyte.

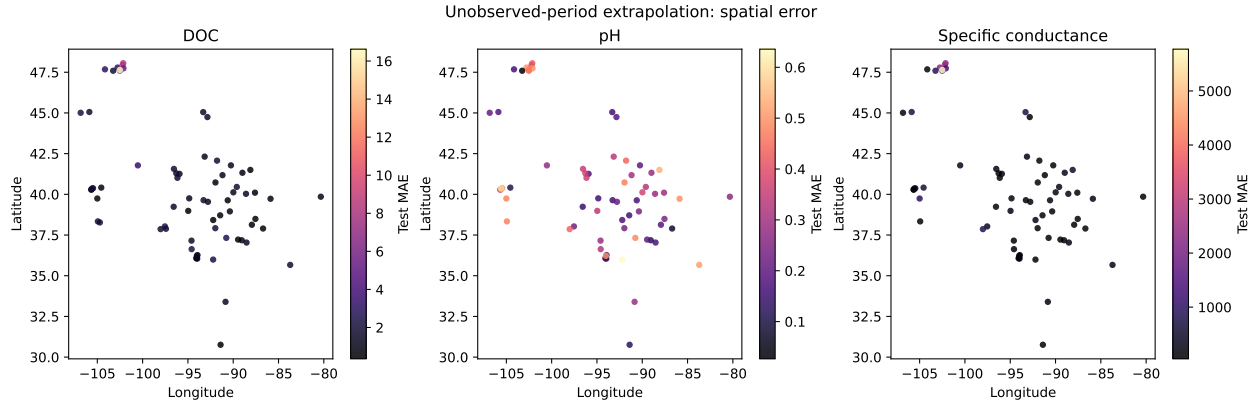


Figure 5: **Spatial error under unobserved-period extrapolation.** Station-level MAE for the five-seed EcoHydroGraph median reconstruction.

4 Discussion

The central finding is that hydro-ecological temporal memory improves sparse water-quality reconstruction, with value determined by the target analyte and missingness process. DOC and specific conductance show large and consistent gains, especially under temporal extrapolation. pH shows smaller gains and near-parity under spatial station holdout. The same model therefore reveals heterogeneity in the recoverable information of water-quality variables.

The graph provides the spatial context in which ecological and hydroclimatic inputs are interpreted. The headline result is the performance pattern created by jointly modeling river structure, ecology, and time. The window curve suggests that a six-month implementation may offer most of the benefit for pH and conductance, while the 12-month configuration remains the best common model. Because the spatial trunk is held fixed, this comparison isolates the temporal extension rather than ranking all possible spatial models.

The primary comparison quantifies improvement over a matched snapshot graph. A separate historical DOC random-forest benchmark performs better on the same query cells (Appendix A), so the present results do not establish superiority over strong tabular learning. This makes a common-protocol comparison with tabular and sequence models the next empirical priority.

The reconstruction products expose where the model is less informative. pH gains are modest, and tail metrics are unstable in scenarios with very few high-value query cells. Ensemble seed spread is a model-dispersion diagnostic rather than a calibrated prediction interval. External-basin replication and a channel-factorial experiment are natural next tests.

5 Conclusion

EcoHydroGraph combines river-network structure, watershed ecology, and hydro-temporal memory for sparse water-quality reconstruction. On ST357, it substantially improves DOC and specific-conductance reconstruction across random, temporal, and spatial missingness, while pH benefits more modestly. The recoverable memory of a river monitoring system is analyte-specific, providing a basis for model-assisted gap filling between direct observations.

Data and code availability

Analysis products, provenance records, and scripts are maintained in the project repository. Main tables are in `experiments/phase4_transfer/temporal_h2x_v1/t8_synthesis/`; five-seed products and figures are in `experiments/phase4_transfer/temporal_h2x_v1/t9_products/`.

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A Historical random-forest context

An audit of the existing DOC ecological random forest verified identical dataset hashes, training/validation/test/context cell sets, and station-month test labels for the four scenarios. Table 4 reports MAE recomputed from the stored predictions for seeds 42–46. The historical forest uses 200 trees, ecological and hydrological covariates, and self-excluded same-month visible-DOC aggregates.

These are historical pilot results with a different model-fitting protocol, included to contextualize the matched graph comparison; they are not a new confirmatory experiment. Existing three-analyte forest experiments use basin-transfer tasks and cannot populate the present four-scenario table.

Table 4: Historical DOC random-forest context on identical query cells. MAE is in mg/L; both columns average five seeds.

Scenario	Query cells	Historical EcoRF	EcoHydroGraph
Random gaps	4,514	1.476	2.625
Unobserved period	2,223	1.186	1.417
Observation-assisted	1,778	1.298	1.413
Unmonitored stations	2,531	2.862	3.876

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